

COMP 4421 (Spring 2026)

Assignment 2

Department of Computer Science and Engineering
The Hong Kong University of Science and Technology

Total = 100 marks

Due: 11:55 pm, April 19, 2026

Assignments must be submitted via CANVAS

Late Policy: Late assignments incur a 10%-score penalty; only one day late is allowed.

Programming Assignment

In this assignment, you should complete the Python functions in the given Jupyter Notebook. Download `programming2.ipynb` and related files from CANVAS.

1. Your submitted Jupyter Notebook must be named `programming2.ipynb`.
2. Compress all files into one file named
[your 8-digit student ID]_assign2.rar (or zip)
Example: 09654321_assign2.rar
3. If multiple submissions are made before the deadline, the latest version will be graded.
4. Plagiarism is taken seriously. You may discuss ideas but **must not share code or pseudocode**.
5. Marks will be deducted for requirement violations.

Grading criteria

- Correctness and efficiency of algorithms.
- Adherence to assignment. 20% penalty for any forbidden extra library imports.

Written Assignment

Submit your written assignment as a PDF named

[your 8-digit student ID]_w-assign2.pdf

Example: 09654321_w-assign2.pdf

Include this file in the same compressed submission.

Question W1: Morphological Opening and Closing

For the following given binary image A :

$$A = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

and structuring elements B and C respectively:

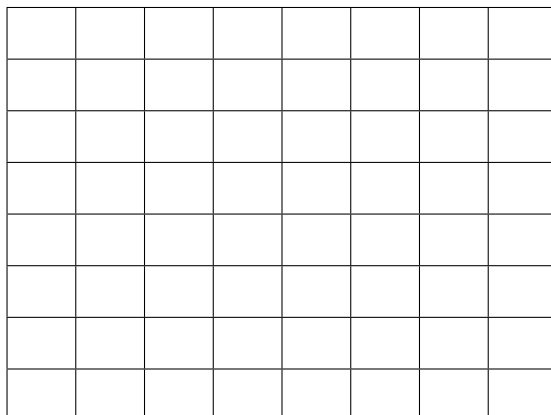
$$B = \begin{array}{|c|c|c|} \hline 0 & 1 & 0 \\ \hline 1 & \underline{1} & 1 \\ \hline 0 & 1 & 0 \\ \hline \end{array}$$

$$C = \begin{array}{|c|c|c|} \hline 0 & 1 & 1 \\ \hline 1 & \underline{1} & 1 \\ \hline 0 & 1 & 0 \\ \hline \end{array}$$

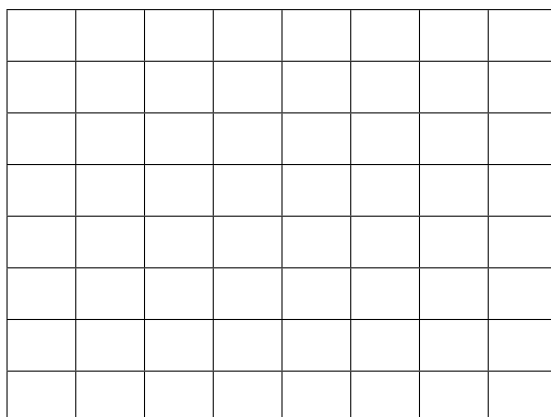
The origin of the SE is underlined as 1.

(i) Calculate the opening $A \circ B$ using the structuring **element** B above.

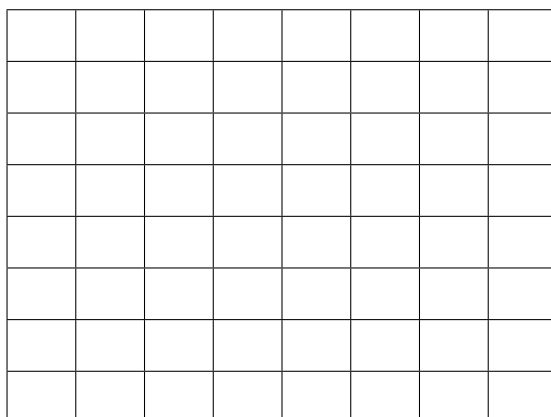
(ii) Calculate the closing $A \bullet C$ using the structuring **element** C above.



(iii) Calculate the operation $A \bullet C \bullet B$.

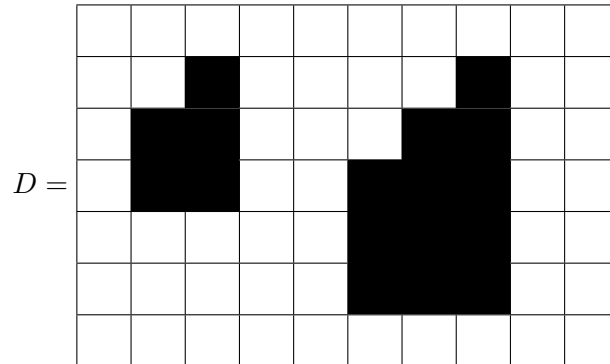


(iv) Calculate the operation $A \bullet C \bullet C$.



Question W2: Morphological Hit-or-Miss Transform

Consider the binary image D , and structuring elements E , F shown below.



$E =$

1	0	1
0	<u>0</u>	1
0	0	1

The origin of SE is underlined as 0.

$F =$

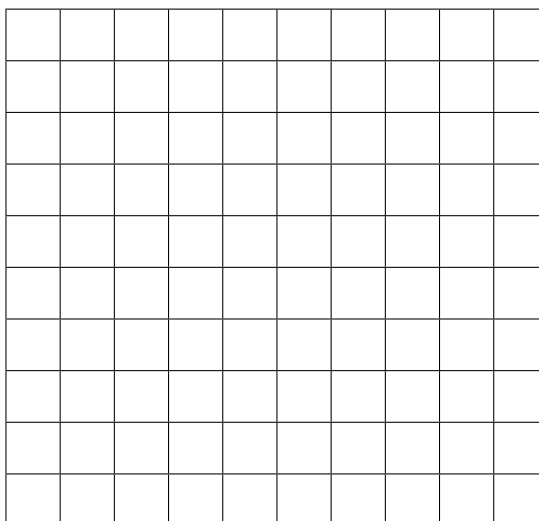
x	x	x
x	<u>0</u>	0
1	1	1

The origin of SE is underlined as 0.

Note: '1' means foreground (white), '0' means background, and 'x' means don't care.

(i) Using the Structuring element E , apply the hit-or-miss transform, indicate which pixels will be detected using $\checkmark$.

Output for $E =$



(ii) Using the Structuring element F , apply the hit-or-miss transform, indicate which pixels will be detected using $\checkmark$.

Output for $F =$

